

SEN381 | SOFTWARE ENGINEERING 381

CivicConnect

Milestone 1 Project Engineering Document

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BASELINE STATUS

Approved Milestone 1 engineering baseline

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0. Document control

This document integrates the team's Milestone 1 foundation, requirements, traceability, risk, forward-engineering, decision and AI-accountability evidence into one controlled Project Engineering Document. Version 1.0 is the team-reviewed and approved Milestone 1 engineering baseline.

Version	Date	Purpose	Owner or approval
0.2	7 September 2026	Member 1 foundation draft	Ryno Goetz
0.3	8 September 2026	Requirements and RTM draft	Steven Riaan Piek
0.9	8 September 2026	Unified M1 review draft with corrected registers	All members
1.0	8 September 2026	Controlled M1 engineering baseline	Approved by Ryno Goetz, Steven Riaan Piek and Willem Booysen

0.1 Baseline approval

All three members reviewed and approved this baseline on 8 September 2026. Gate outcome: ACCEPTED. The requirements, traceability, risks, decisions, governance controls and AI Usage Register were reviewed as part of that approval. Open client-policy questions remain controlled deferrals and do not prevent the M1 baseline.

1. Problem and business need

1.1 Problem statement

CivicConnect addresses a loss of control over the lifecycle of service requests. The organisation receives operational requests through several informal and disconnected channels, but those channels do not provide one dependable record connecting a request to its owner, current state and actions taken. A request can therefore reach someone without becoming visible work that can be assigned, followed up and verified as completed [S4, section 2].

This fragmentation affects three groups differently. Requesters cannot reliably determine what happened after submission. Staff must reconcile information and ownership across channels while organising work. Management cannot confidently distinguish outstanding work from completed work or reconstruct why a request changed state. Inconsistent handling of sensitive request information compounds the visibility and accountability problem [S4, sections 2-2.1].

The underlying engineering problem is consequently more than collecting request descriptions: the platform must preserve useful information and controlled responsibility from submission through feedback and authorised closure. Improving the interface alone would not establish that control if assignment, permitted status changes, access boundaries and reporting definitions remained ambiguous. This is analysis derived from the scenario, not an observation from a conducted stakeholder interview.

1.2 Causes, effects and intended response

Request information is spread across messages, calls, spreadsheets and paper

Work may be missed or duplicated; reconciliation consumes staff effort.

Proposed engineering response: Maintain a retrievable digital request record with controlled classification and clear submission outcome. **Stakeholder/need link:** STK-001, STK-002; NEED-001, NEED-002.

Ownership and progress are difficult to see

Requesters repeatedly seek updates; staff may work on the same request or leave it unowned.

Proposed engineering response: Expose authorised status/history, explicit responsibility and lifecycle feedback. **Stakeholder/need link:** STK-001, STK-002, STK-004; NEED-003 to NEED-005.

Status changes and actions lack dependable accountability

A completion claim is difficult to explain or check.

Proposed engineering response: Control state changes and retain attributable action/resolution information. **Stakeholder/need link:** STK-002, STK-003; NEED-006, NEED-008.

Management reporting is manual and inconsistent

Workload and overdue work cannot be assessed consistently.

Proposed engineering response: Derive oversight views from request records using agreed definitions and useful category/status dimensions. **Stakeholder/need link:** STK-003; NEED-007, NEED-008.

Sensitive information is exchanged inconsistently

Visibility can expose details to people who do not need them.

Proposed engineering response: Define role-related access and sensitive-data handling early; later verify access with positive and negative tests. **Stakeholder/need link:** STK-001, STK-002, STK-005; NEED-009.

The scenario problems are grounded in the scenario [S4, section 2]. Consequences and responses are the team's proposed analysis. They do not imply that message import, automatic duplicate detection or a particular authentication technology is required.

1.3 Business need

The organisation needs an affordable, supportable platform that lets requesters submit and follow requests, lets authorised staff take responsibility and manage them, and gives management a dependable view of service activity. Its value depends on the same request information supporting operational work and oversight, with access and actions controlled throughout the lifecycle [S4, sections 2.1 and 3].

The proposed product scope therefore prioritises the required request-management workflow and sufficient accountability. Automatic imports from existing channels, payments and predictive routing are excluded or deferred as proposals because they would create additional integration, security, testing and cost obligations without being necessary to demonstrate the required minimum capabilities [S4, section 3.1; sections 3.4-3.5].

1.4 Intended value and how to assess it

The following evaluation methods turn business value into checkable outcomes. They are not measured implementation results. The requirements section translates them into requirements and acceptance criteria using the approved M1 working policies and measurable baseline targets [S4, sections 5, 11 and 15].

VAL-001

A successfully submitted request remains findable.

Proposed validation evidence: Submit an agreed set of valid requests and check that each acknowledged request can subsequently be retrieved by its authorised requester.

Scope/need link: SCP-001, SCP-004; NEED-001, NEED-003.

VAL-002

Requesters can see meaningful progress.

Proposed validation evidence: Walk through accepted, rejected, updated and completed outcomes; verify that feedback and the visible state agree with the recorded outcome.

Scope/need link: SCP-003, SCP-005; NEED-003.

VAL-003

Staff can identify responsibility and permitted next actions.

Proposed validation evidence: Exercise assignment/acceptance and authorised state changes; verify ownership, retained action information and rejection of unauthorised changes. **Scope/need link:** SCP-009 to SCP-012, SCP-017; NEED-005, NEED-006, NEED-009.

VAL-004

Management can rely on its work overview.

Proposed validation evidence: Use a known set of requests with agreed overdue rules and compare open/overdue/resolved/closed views and category/status breakdowns with expected records. **Scope/need link:** SCP-013 to SCP-016; NEED-007, NEED-008.

VAL-005

Better visibility does not expose restricted information.

Proposed validation evidence: Verify requester and staff visibility against the agreed permission matrix, including attempts to access another user's restricted request.

Scope/need link: SCP-006, SCP-017; NEED-009.

VAL-006

The solution remains workable within team and operating constraints.

Proposed validation evidence: Review actual scope delivery, schedule variance, quality/security evidence, service limits and likely running cost at later milestones.

Scope/need link: CON-001 to CON-010; NEED-010, NEED-011.

No percentage reduction in lost requests, turnaround time or operating cost is claimed. The brief provides no measured current-process baseline. Such claims would require a defined before/after comparison and trustworthy measurements.

1.5 Boundaries and related requirements

The initial category set is Maintenance, IT support, Facility fault, Damaged equipment, Security concern, Lost property and Other. A request records a title, description, category, location, requester identity/contact details, automatic submission date/time, a sensitive-information indicator and an optional attachment. The software does not itself repair equipment or provide physical emergency response. The team should validate this working taxonomy during requirements review and control later changes.

The requirements section uses the stakeholder needs and proposed success evidence as inputs. The risk and forward-engineering sections address the risks implied by lost ownership, inconsistent reporting and sensitive-data access. The team should judge success using stakeholder value, controlled scope, time, resources, quality, security and operational evidence, rather than a successful demonstration alone [S4, sections 5 and 21].

References: [Source register](#), especially S4 sections 2-5, 11, 15 and 21; S1 section 3.

2. Stakeholder analysis

2.1 Method and evidence limits

This is a scenario-based analysis. Requester, staff and management roles come directly from the Master brief [S4, section 3]. Coordination, operational support and assessment roles below are analytical groupings inferred from the described responsibilities and lifecycle controls. They do not assert that separate departments or named client representatives exist. One person may perform several organisational roles; the final permission model must distinguish responsibilities rather than assume one job title equals one software role.

Influence means ability to affect scope, priorities, acceptance or policy. Interest means how directly the stakeholder is affected by the system's outcomes. High/medium ratings are qualitative judgements for engagement planning, not measured survey results. All ratings and conflicts require review.

2.2 Stakeholder register

STK-001

Requesters; explicit in S4 section 3.

Needs and expectations: Submit understandable, appropriately classified requests; see their status/history and meaningful feedback; protect sensitive details. **Influence and interest:** Medium / high: directly affected, but no policy authority is stated.

Proposed engagement: Validate submission and feedback scenarios; ask what acknowledgement and progress information are useful.

STK-002

Authorised service staff; explicit in S4 section 3.

Needs and expectations: Find relevant work and details, take responsibility, apply permitted state changes and record actions/resolutions. **Influence and interest:** High / high: workflow feasibility and data quality depend on staff use.

Proposed engagement: Walk through assignment, reassignment, rejection, resolution and closure cases; verify access and required information.

STK-003

Management/oversight; explicit in S4 section 3.

Needs and expectations: Dependable activity views, identification of overdue/open/resolved/closed work, accountability and useful analysis. **Influence and interest:** High / high: influences priorities, reporting definitions and acceptance.

Proposed engagement: Agree reporting meaning and visibility; validate summaries against a known request set.

STK-004

Service coordination responsibility; inferred from assignment/prioritisation problems in S4 section 2; may be held by STK-002 or STK-003.

Needs and expectations: Clear responsibility, controlled categories and consistent handling of unassigned or incorrectly routed work. **Influence and interest:** High / high for workflow policy.

Proposed engagement: Clarify who may assign, change category, reassign or close work; validate boundary cases.

STK-005

Organisation's operational/security support responsibility; inferred from S4 sections 16-18; named owner unconfirmed.

Needs and expectations: Appropriate access, manageable configuration, protected credentials, recoverability and sustainable operation. **Influence and interest:** Medium / high; actual authority must be confirmed.

Proposed engagement: Review sensitive-data questions and later deployment/support constraints without choosing a stack in M1.

STK-006

Project team: Ryno, Steven and Willem; S2 and S4 sections 7.1-8.

Needs and expectations: Clear scope and dependencies, feasible workload, shared understanding and maintainable engineering evidence. **Influence and interest:** High / high for engineering choices within the brief.

Proposed engagement: Daily WhatsApp/Discord discussions, meetings every second day after 11:00, issues, meaningful two-reviewer PRs and joint baseline review.

STK-007

SEN381 lecturer/assessor; S4 sections 19, 20 and 23.

Needs and expectations: A defensible product and authentic individual/team evidence that meets the project controls. **Influence and interest:** High / high for assessment; not assumed to be the real organisation's management.

Proposed engagement: Clarify ambiguous assessment requirements and demonstrate controlled evidence at milestones.

2.3 Stakeholder needs for requirements derivation

Need ID	Need	Stakeholder/source	Priority basis and scope link
NEED-001	A valid new request can be recorded with enough information to act on it	STK-001, STK-002; S4 section 3, requester submission	Required minimum; SCP-001
NEED-002	Requests use a controlled category mechanism that supports routing and reporting	STK-001, STK-004; S4 section 3, requester categorisation	Required minimum; SCP-002

Need ID	Need	Stakeholder/source	Priority basis and scope link
NEED-003	Requesters can find their earlier requests, inspect current status and understand submission/lifecycle outcomes	STK-001; S4 section 3, requester visibility and feedback	Required minimum; SCP-003 to SCP-005
NEED-004	Staff can locate relevant authorised work and inspect its details using useful search/filter/sort criteria	STK-002; S4 section 3, staff visibility/search/details	Required minimum; SCP-006 to SCP-008
NEED-005	A request has explicit responsibility that can be assigned or accepted under agreed permissions	STK-002, STK-004; S4 sections 2 and 3	Required minimum; SCP-009
NEED-006	Changes, actions and resolution/closure follow defined responsibilities and permitted transitions	STK-002, STK-004; S4 section 3, staff lifecycle and actions	Required minimum; SCP-010 to SCP-012
NEED-007	Management can identify service activity and distinguish open, overdue, resolved and closed work	STK-003; S4 section 3, management oversight	Required minimum; SCP-013, SCP-014
NEED-008	Management can analyse work by useful dimensions and obtain sufficient information to explain responsibility and progress	STK-003; S4 section 3, management analysis/accountability	Required minimum; SCP-015, SCP-016
NEED-009	Access and disclosure match justified responsibilities, including treatment of sensitive request information	STK-001, STK-002, STK-005; S4 sections 2, 3 and 16	Cross-cutting security obligation; SCP-006, SCP-017
NEED-010	The platform can be verified, operated and recovered using available resources	STK-005, STK-006; S4 sections 4, 15, 17 and 18	Lifecycle obligation; SCP-018; CON-002, CON-009, CON-012
NEED-011	The team can demonstrate controlled collaboration and defend evidence individually	STK-006, STK-007; S4 sections 8-11 and 19	Project/process obligation; CON-001, CON-007, CON-008

These needs are not the final functional/non-functional requirements. Steven must add unique requirement IDs, priority, precise behaviour and measurable acceptance criteria and link them through the RTM [S4, section 11].

2.4 Meaningful conflicts and proposed resolutions

Visibility versus confidentiality

Requesters and management want useful progress information; staff/support must prevent inappropriate disclosure.

Approved resolution: Separate requester-visible progress from restricted details and apply the least-privilege baseline. Consequence for later work: M2 must refine the permission matrix for UI/API access checks, data exposure rules and negative security tests; this does not justify an unrestricted management role.

Quick submission versus actionable information

Requesters benefit from a short form; staff need enough context to categorise and act.

Proposed resolution for validation: Agree only the necessary required fields, validate them clearly, and record how missing information is handled. **Consequence for later work:** Affects validation, acceptance criteria, data collection and usability; field list remains Q-001.

Flexible handling versus accountable status

Staff need to correct assignments and update work efficiently; management needs consistent history.

Proposed resolution for validation: Define authorised transitions and retain attributable changes; decide how correction, rejection and closure are represented. **Consequence for later work:** Shapes lifecycle requirements, reporting definitions and tests; exact transitions remain Q-003.

Rich reports versus sustainable scope

Management could benefit from exports and predictive analysis; the three-person team has limited capacity.

Proposed resolution for validation: Deliver the mandatory oversight views first; propose advanced analytics as future scope subject to impact analysis. **Consequence for later work:** Preserves required reporting while controlling extra security, implementation, testing and cost obligations.

Fast completion versus independent review

Ryno wants to finish promptly; team governance requires both other members to review substantive changes.

Proposed resolution for validation: Submit coherent artefact units early and reserve review time within daily capacity. **Consequence for later work:** Requires authentic PR/review evidence and prevents approval becoming a last-minute dependency.

These conflicts are reasoned from the scenario and project constraints, not claims of observed stakeholder disputes.

2.5 Requirements traceability examples

Example A - requester visibility. S4 section 3 -> STK-001 -> NEED-003 -> SCP-003/SCP-004. Candidate behaviour: an authorised requester can retrieve the current status and history/list of their submitted requests. Candidate acceptance scenario: seed two requesters with distinct requests; verify each sees the correct own list and current state after a permitted staff update. Whether either may see another person's requests must follow the agreed access policy, with denied-access cases included. Final FR/NFR and acceptance IDs are recorded in the requirements and RTM sections.

Example B - overdue reporting. S4 section 3 -> STK-003 -> NEED-007 -> SCP-014.

Working behaviour: a request is overdue when its agreed target date/time has passed and it is not Resolved, Closed or Rejected. The target is set by authorised staff or a manager during acceptance/assignment. Candidate acceptance scenario: use records either side of the due-time boundary and records in excluded states; compare the result to this rule. Steven should convert the rule into a testable requirement and the team should formally approve it.

These examples demonstrate currently available traceability. Design, implementation, test execution and release evidence are added later, not marked complete now [S1, section 3; S4, section 11.1].

References: [Source register](#), S4 sections 2-4, 8, 11, 16-19; S1 section 3; S2 and S3.

3. Scope baseline

3.1 Scope objective and system boundary

CivicConnect will support the digital management of service requests from submission and classification through authorised staff handling, feedback and management oversight. The boundary includes request records, responsibility, state changes, relevant action/resolution information, role-appropriate views and meaningful service reporting. Physical fulfilment of the requested work remains an organisational activity outside the software [S4, sections 2-3].

The initial product serves one organisation. The scope below carries every minimum capability in the Master brief into the approved M1 baseline and records the working policies needed for requirements engineering. It defines the agreed scope but does not claim completed implementation [S4, sections 3 and 11; S1, section 3].

3.1.1 Selected request and workflow policy

- Required request information: title, description, category, location, requester identity/contact, automatically recorded submission date/time and a sensitive-information indicator; an attachment is optional.
- Initial categories: Maintenance, IT support, Facility fault, Damaged equipment, Security concern, Lost property and Other.
- Lifecycle: Submitted -> Accepted -> Assigned -> In Progress -> Resolved -> Closed. Submitted requests may be Rejected; authorised users may reassign; a requester or manager may request reopening; every status change must be attributable.
- Overdue rule: the agreed target date/time has passed and the request is not Resolved, Closed or Rejected. Authorised staff or a manager sets the target on acceptance or assignment.
- Feedback: show in-application feedback for accepted, rejected, updated, resolved and closed requests. External delivery channels are deferred.

3.2 In-scope minimum business capabilities

All items SCP-001 to SCP-016 represent required minimum capabilities from S4 section 3. Their detailed behaviour, fields, policies and verification conditions are linked to Steven's requirements and acceptance criteria. "Required" is the priority basis from the brief; these items are baselined in PED v1.0.

Scope ID	Included capability	Stakeholder/need	Boundary and clarification
SCP-001	Submit a new service request with appropriate information	STK-001; NEED-001	Use the selected fields in section 3.1.1; no automatic import from existing channels is assumed
SCP-002	Categorise a request through a controlled mechanism	STK-001, STK-004; NEED-002	Use the initial category set in section 3.1.1; controlled changes remain possible; automated AI classification is not required

Scope ID	Included capability	Stakeholder/need	Boundary and clarification
SCP-003	View the current status of submitted requests	STK-001; NEED-003	Apply the agreed visibility policy; make the status meaningful to requesters
SCP-004	View a history/list of previously submitted requests	STK-001; NEED-003	Preserve retrievability under the agreed retention policy; not a promise of indefinite storage
SCP-005	Receive meaningful feedback on accepted, rejected, updated and completed requests	STK-001; NEED-003	Provide in-application feedback for accepted, rejected, updated, resolved and closed events; external channels are deferred
SCP-006	View work relevant to authorised staff	STK-002; NEED-004, NEED-009	Define relevant/authorised visibility; staff access must not be assumed universal
SCP-007	Search, filter or sort requests using useful criteria	STK-002; NEED-004	Agree useful criteria with staff; the brief does not prescribe an advanced search engine
SCP-008	View full request details when authorised	STK-002; NEED-004	Full details are permission-dependent, including sensitive information
SCP-009	Assign or accept responsibility for a request	STK-002, STK-004; NEED-005	Authorised staff may accept/assign work; a coordinator or manager may assign and reassign it
SCP-010	Update status through controlled transitions	STK-002, STK-004; NEED-006	Apply the lifecycle and attributable-change rule in section 3.1.1
SCP-011	Record relevant actions, comments or resolution information	STK-002; NEED-006	Define required accountability and requester-visible versus restricted information
SCP-012	Resolve or close requests where authorised	STK-002, STK-004; NEED-006	Clarify resolved versus closed, rejection and any reopening policy
SCP-013	View useful service activity information	STK-003; NEED-007	Agree an oversight view suited to the organisation's decisions
SCP-014	Identify open, overdue, resolved and closed work	STK-003; NEED-007	Apply the overdue definition in section 3.1.1 and verify its time boundary

Scope ID	Included capability	Stakeholder/need	Boundary and clarification
SCP-015	View requests by category/status or other justified dimensions	STK-003; NEED-008	Core category/status visibility is included; additional dimensions require value justification
SCP-016	Access enough information for accountability and service-performance analysis	STK-003; NEED-008	Retain sufficient attributable request history; this does not require predictive analytics or a data warehouse

3.3 Cross-cutting obligations

Scope ID	Included obligation	Source and implication
SCP-017	Appropriate access and sensitive-data protection throughout the request lifecycle	S4 sections 2, 4 and 16; supports NEED-009. Define role/access and disclosure rules now; defer the authentication technology choice, not the security requirements
SCP-018	Measurable quality, verification and controlled delivery/operation across later milestones	S4 sections 15-18; supports NEED-010. M1 identifies concerns/requirements; design, tests, staging, release, recovery and operational evidence develop through later milestones

These obligations remain part of the overall project even though their full implementation is not an M1 deliverable.

The access baseline applies least privilege: requesters submit and view their own requests; authorised staff view relevant work and perform permitted updates; coordinators/managers assign, reassign and view reporting; management sees authorised summaries/details without blanket access to sensitive information; and administration rights do not automatically grant business-data access. Use fictional, non-sensitive test data. The exact retention period will be confirmed later because neither supplied brief fixes one.

3.4 Recommended out-of-scope items for the initial product baseline

The following are recommended for team sign-off. They are not exclusions stated by the lecturer and must not remove a capability required above.

OUT-001

Live two-way ingestion/synchronisation with WhatsApp, email, telephone systems or paper workflows.

Justification: The existing channels explain fragmentation; S4 section 3 does not require integrations. Supporting several channels adds reconciliation, privacy, credentials, dependency and testing obligations. **Effect and revisit condition:** Requests are entered through CivicConnect under SCP-001. Revisit if validated stakeholder evidence makes integration necessary and the team approves its impact.

OUT-002

Billing, payments and procurement/inventory management.

Justification: They are separate business processes beyond the stated request lifecycle and add security and reconciliation work. **Effect and revisit condition:** Record/request operational work without implementing financial or stock-management systems. Revisit only through justified scope change.

OUT-003

Performing repairs, emergency dispatch or guaranteeing real-world service completion times.

Justification: The software records and coordinates work; physical fulfilment and emergency response require organisational resources beyond the software boundary. **Effect and revisit condition:** Security-related concerns may still be recorded. Validate user-facing escalation expectations; do not market the platform as an emergency-response service.

Defence of OUT-001: a messaging integration could reduce repeated entry, but it must also resolve identity, duplicate submissions, update conflicts, access and delivery failures. The required business value can first be demonstrated through controlled submission and tracking inside CivicConnect. Proposing this exclusion preserves all minimum capabilities while keeping work compatible with three members' capacity. Its trade-off is an adoption/manual-entry burden, which should be recorded as a risk for Willem to assess rather than ignored [S4, sections 3.1, 4, 12 and 18].

3.5 Recommended future or deliberately deferred work

FUT-001

Native mobile applications and offline synchronisation.

Why it is not a current commitment: These create additional platforms, synchronisation states and verification effort beyond the minimum capabilities. **Information required before deciding:** User/device/connectivity needs, accessibility evidence, schedule and maintenance impact.

FUT-002

AI routing, duplicate detection and predictive service analytics.

Why it is not a current commitment: Basic controlled categories and oversight can satisfy the minimum capabilities; automated inference adds accuracy and accountability questions. **Information required before deciding:** Evidence of value, suitable data, failure handling, privacy and operational-cost analysis.

FUT-003

Bulk migration of historic email, spreadsheets and paper records.

Why it is not a current commitment: Legacy sources are fragmented and their formats, quality and access permissions are not supplied. **Information required before deciding:** Sample formats, volume, reconciliation/retention policy and a justified migration requirement.

FUT-004

External message delivery such as SMS, WhatsApp or email notifications.

Why it is not a current commitment: Required feedback under SCP-005 remains included; external channels are not yet justified. **Information required before deciding:** Agreed feedback timing/channel needs, delivery failure handling, service limits, cost and sensitive-content policy.

FUT-005

Final technology stack, architecture and detailed persistence/UI/API design.

Why it is not a current commitment: These are later lifecycle decisions, explicitly outside required M1 decisions/implementation. **Information required before deciding:** Baseline FRs/NFRs, constraints, team capability, environment compatibility and comparison evidence.

In-application feedback is the selected M1 working policy for SCP-005. Deferring an external channel must never silently remove meaningful feedback. Likewise, FUT-003 does not remove required history for requests submitted to the new platform.

Willem should record only genuine team decisions or justified deferments in the Decision Log after review; these proposals are decision inputs, not approved entries [S4, section 13].

3.6 M1 milestone boundary

M1 is due on 9 September 2026 and produces the controlled problem, stakeholder, scope, requirements, constraints, traceability, risk, forward-thinking, decision and governance foundation in PED v1.0. Final stack/architecture selection, detailed database/UI/API or design-pattern implementation, CI pipeline implementation, extensive application code and production deployment are not required M1 deliverables unless approved as exploration [S1, sections 3-5; S3].

This separates when the team performs work from what the final product must support. Security, testability, deployment and recovery must be considered now even though implementation occurs later.

3.7 Approval and controlled change

Steven checked every SCP item and selected policy against the requirements and acceptance criteria and linked source and stakeholder needs through the RTM. Willem reviewed the risks, forward concerns and decisions. All three members approved the exclusions, controlled deferrals and unified PED v1.0 baseline [S4, section 11 and Appendix D].

After approval, changes require a recorded request and impact analysis, a decision and authorisation, controlled implementation and verification, then updates to the affected baseline artefacts. Analyse scope, schedule/resources, cost, quality/security, requirements, design/data/interfaces, tests, deployment and risks as applicable; use S4 Appendix E rather than silently absorbing additions [S4, section 14].

References: [Source register](#), S4 sections 2-4, 11-18 and Appendices D-E; S1 sections 3-5.

4. Constraints and assumptions

4.1 Constraint register

The Master brief defines constraints on scope, four-milestone delivery, cost, quality, security and technology decisions. The following analysis applies those constraints to this team and CivicConnect's request workflow [S4, sections 4 and 18]. Proposed IDs support traceability and do not replace Willem's Risk Register.

CON-001

Exactly three team members; each must contribute and defend the full project.

Source/status: S4 sections 7.1-8; S2 records Ryno, Steven and Willem. **Engineering implication and planned response:** Assign primary owners and both non-author reviewers. Plan review capacity; specialisation does not remove shared understanding.

CON-002

All three members are available every day for four hours each.

Source/status: S3; confirmed by Ryno. **Engineering implication and planned response:** Gross daily capacity is 12 person-hours. Discussion, meetings, review and corrections count within it; record actual effort separately.

CON-003

Delivery through four formal milestones; M1 is due 9 September 2026.

Source/status: S4 sections 4 and 20; S3 confirms the M1 date. **Engineering implication and planned response:** Complete author checks and handoff early enough for two genuine non-author reviews; later milestone dates still need the formal schedule.

CON-004

WhatsApp/Discord, daily discussions and a meeting every second day after 11:00.

Source/status: S3; working arrangement agreed by all three members. **Engineering implication and planned response:** Use frequent updates for dependencies; transfer material decisions/actions to controlled repository records.

CON-005

Every minimum business capability must remain in controlled product scope; additions need justification.

Source/status: S4 sections 3-4; SCP-001 to SCP-016 map the minimums. **Engineering implication and planned response:** Protect the core lifecycle and oversight. Assess integrations/analytics separately; do not silently drop feedback, overdue views or authorised closure.

CON-006

Requirements and quality expectations need traceable, measurable evidence.

Source/status: S4 sections 11 and 15; S1 section 3. **Engineering implication and planned response:** Define useful performance/usability/reliability criteria and test conditions with Steven. Avoid unsupported adjectives or invented traffic targets.

CON-007

One controlled repository, protected main, PRs and two independent non-author approvals; no substantive direct development on main.

Source/status: S4 section 9; all three members have full repository access, but protection/evidence must be checked. **Engineering implication and planned response:** Use task branches and meaningful reviews for documents as well as code. A one-approval setting alone would not meet the brief's two-approval rule.

CON-008

PED/RTM evolve with controlled history; evidence and AI accountability must be authentic.

Source/status: S4 sections 6, 8, 10-11 and 23. **Engineering implication and planned response:** Preserve identifiers/version history, record real contributions and verification, and integrate approved Member 1 content into the PED.

CON-009

Prefer free/low-cost services where practical while recognising limits and likely ongoing costs.

Source/status: S4 sections 4 and 18; Ryno reported no additional budget, device, connectivity or software limitations. **Engineering implication and planned response:** Evaluate future service quotas, storage, messaging and operating cost. Optional external notifications may affect both budget and reliability.

CON-010

Request data may be sensitive; security is required throughout the lifecycle.

Source/status: S4 sections 2, 4 and 16. **Engineering implication and planned response:** Define authorised visibility, least privilege, permitted status actions and sensitive-information handling. Verify denial as well as authorised use later; record residual risk.

CON-011

No prescribed stack; the team must justify capability and environment compatibility.

Source/status: S4 sections 4, 18.1 and 25. **Engineering implication and planned response:** Do not choose a stack in M1 without evidence. Assess learning curve, available devices/environment, security, testing and hosting fit before a later decision.

CON-012

Development, test, staging and production have distinct verification and operational concerns.

Source/status: S4 section 17. **Engineering implication and planned response:** Identify configuration, recovery and operational evidence needs now, for Willem's forward considerations. M1 does not implement production deployment.

4.2 Assumptions requiring validation

ASM-001

One organisation's service-request workflow is the selected initial context; no multi-tenant product commitment.

Impact if wrong: Multiple independent organisations could alter identity, isolation, reporting, costs and administration. **Owner/action:** Team confirms during formal scope review.

ASM-002

In-application feedback is the selected initial mechanism; external messaging is deferred.

Impact if wrong: A later off-platform need could require delivery retries, privacy controls and added costs. **Owner/action:** Steven specifies in-app outcomes; Willem records the deferment/risk.

ASM-003

Coordination/management is a responsibility set rather than proof of a separate organisational department.

Impact if wrong: A real distinct role could require adjusted permissions and workflow cases. **Owner/action:** Validate when an organisational representative becomes available.

ASM-004

Four hours per member per day includes meetings and reviews, and all members are available daily.

Impact if wrong: Confirmed by Ryno on 7 September 2026. **Owner/action:** Plan and record effort on this basis; update if availability changes.

ASM-005

Fictional, non-sensitive test requests will be used for early verification.

Impact if wrong: A later need for real records would introduce approval, access and data-handling constraints. **Owner/action:** Steven/Willem preserve this test-data rule and record any exception.

Except for ASM-004, these are team-selected planning positions rather than confirmed client facts. Material assumptions should be reviewed for risk treatment under S4 section 12.

4.3 Constraint interactions and trade-offs

Visibility versus sensitive information (CON-005/CON-010). SCP-003 and SCP-008 require meaningful views, while the scenario includes sensitive requests. Showing every field to every user would simplify a display but violate role-related access needs. Define requester-visible progress separately from restricted information and then derive a permission matrix. This affects requirements, data/API/UI boundaries, audit information and negative tests; the actual authentication technology remains a later decision.

Fast completion versus review capacity (CON-001/CON-002/CON-007). Each author needs the other two members' approvals. Completing Ryno's drafts early cannot by itself complete the baseline if Steven or Willem is unavailable. Reserve daily review time and resolve cross-artefact questions at the every-second-day meeting. Maintain meaningful review instead of treating approval as administrative overhead.

Feedback versus cost and delivery reliability (CON-005/CON-006/CON-009). Feedback is required. The working baseline uses in-application feedback and defers external notification services, which would add delivery failure states, usage limits, configuration and likely operating costs. If later evidence requires external delivery, assess scope, reliability, security, tests and budget together.

Overdue reporting versus workflow policy (CON-005/CON-006). The selected rule marks work overdue when its agreed target date/time has passed and it is not Resolved, Closed or Rejected; authorised staff or a manager sets the target during acceptance/assignment. Steven must make the time boundary testable and link the outcome to SCP-010/SCP-012/SCP-014 and the RTM.

Technology familiarity versus future operation (CON-003/CON-009/CON-011/CON-012).

A familiar development tool may reduce initial effort but still create compatibility, hosting, recovery or service-cost problems. Keep stack selection open in M1 and gather the evidence needed for M2. The institution does not guarantee that every chosen technology will run on its environment [S4, sections 18.1 and 25].

4.4 Policy answers and remaining questions

The team can use the following working answers now and validate them with an appropriate lecturer/client representative when required.

Q-001

Use the request fields and seven categories in scope section 3.1.1; category changes are controlled by authorised management/administration.

Affected work: SCP-001/SCP-002; validation, classification and quality criteria.

Responsible follow-up: Steven converts to requirements; team reviews.

Q-002

Requesters see their own work; authorised staff see relevant work; coordinators/managers assign and report; management has no blanket sensitive access; administration does not imply business-data access.

Affected work: SCP-006/SCP-008 to SCP-012/SCP-016/SCP-017. **Responsible follow-up:** Steven specifies permission cases; Willem reviews security risk.

Q-003

Submitted -> Accepted -> Assigned -> In Progress -> Resolved -> Closed, with authorised rejection, reassignment and requested reopening; all changes attributable.

Affected work: SCP-003/SCP-005/SCP-010/SCP-012/SCP-014. **Responsible follow-up:** Steven specifies transitions and acceptance cases.

Q-004

Overdue means the target date/time has passed and state is not Resolved, Closed or Rejected; authorised staff/manager sets target on acceptance/assignment.

Affected work: SCP-014 and reporting acceptance cases. **Responsible follow-up:** Steven makes the boundary testable.

Q-005

Provide in-application feedback for accepted, rejected, updated, resolved and closed events; external channels are deferred. Exact response-time targets remain to be agreed.

Affected work: SCP-005, FUT-004, cost/reliability/security. **Responsible follow-up:** Steven specifies behaviour; Willem records deferment.

Q-006

Apply least privilege, a sensitive-information indicator and fictional/non-sensitive test data. Exact retention period remains open because neither brief states it.

Affected work: SCP-004/SCP-008/SCP-016/SCP-017 and recovery concerns.

Responsible follow-up: Steven/Willem preserve the open retention item.

Q-007

No extra budget/device/connectivity/software limitations are reported. Exact user volumes, operating hours and measurable performance targets remain open.

Affected work: Performance, usability, reliability and operational requirements.

Responsible follow-up: Steven proposes measurable targets for team review.

The M1 submission date is 9 September 2026. All three members have full repository access and no extra resource limitations are reported. Team-registration evidence remains unconfirmed; the Master lists 31 August 2026 as the registration deadline [S4, section 7.1]. No missed registration is inferred from the absence of evidence. No separate M1 Excel Rubric/Marking Guide has been received.

4.5 Links to requirements and risks

Steven links constraints, assumptions and policy answers to requirements and the RTM. Willem assesses risks such as review delay, ambiguous overdue policy, inappropriate data visibility and future service costs, using his required probability/impact/mitigation fields. When an assumption is validated or rejected, update this record and affected scope/requirements/risks under review; after baseline apply formal change control [S4, sections 11-14].

References: [Source register](#), S4 sections 2-4, 6-18, 20, 23 and 25; S1 sections 3-5; S2 and S3.

5. Team Working Agreement

5.1 Purpose and responsibilities

This agreement applies the Master brief's collaboration and accountability requirements to the team's reported arrangements [S4, sections 7.1-10; S3]. The role allocation is recorded in the repository README [S2]. Ryno confirmed on 7 September 2026 that Steven and Willem agreed to these responsibilities and working arrangements.

Member	Primary M1 responsibilities	Required non-author reviewers
Ryno Goetz	Problem/business need, stakeholder analysis, scope, constraints, working agreement and GitHub governance	Steven Riaan Piek and Willem Booysen
Steven Riaan Piek	Functional/non-functional requirements, acceptance criteria and Requirements Traceability Matrix (RTM)	Ryno Goetz and Willem Booysen
Willem Booysen	Risk Register, forward considerations, Engineering Decision Log, AI Usage Register and PED integration/sign-off coordination	Ryno Goetz and Steven Riaan Piek

Each owner prepares source-supported work, responds to review and maintains its connections with other artefacts. Every member must understand the whole project, participate in presentation/defence and contribute authentic issue, commit, PR, review and documentation evidence [S4, sections 8-8.1 and 19].

5.2 Communication and capacity

- **WhatsApp:** agreed channel for short progress messages, availability and blockers.
- **Discord:** agreed channel for detailed discussions, screen sharing and scheduled meetings.
- **Daily discussions:** each member reports what changed, what comes next, dependencies and any blocked work. Summarise material decisions/actions in the relevant issue or controlled log.
- **Meetings every second day after 11:00:** inspect progress and linked artefacts, resolve policy/scope/requirement conflicts, plan reviews and record action owners. Confirm the exact start time for each meeting through the team channels.
- **Four hours per member every day:** confirmed by Ryno. Discussion, meetings, research, writing, review and corrections all count within this allowance, giving a gross planned capacity of 12 person-hours per day.

The team's WhatsApp/Discord use is for project coordination. It does not commit CivicConnect to integrating those platforms as product features.

5.3 Work and review cycle

1. Create/select an issue for each coherent work unit, with an owner, intended output, dependencies and completion criteria. Work backward from the M1 submission date of 9 September 2026 and leave time for both required reviews.

2. Work on a task branch from the latest controlled main state. Substantive direct development on main is not permitted [S4, section 9].
3. Verify sources, assumptions, identifiers, cross-links and acceptance implications before requesting review. Record material AI assistance and the human verification actually performed.
4. Open a PR linked to the issue and request both non-author members' reviews. Summarise the proposed change, rationale, affected artefacts and unresolved questions.
5. Reviewers inspect substance, including correctness, scope/requirement alignment, security/privacy, traceability and downstream verification implications. Give specific findings where correction is needed.
6. The author addresses comments, reviewers inspect corrections, and the final change receives both required independent approvals before merge. Self-approval and unexamined approval are not accepted [S4, sections 9-9.1].
7. Update related artefacts and the issue after the controlled change. Preserve the review and version evidence.

A drafted file is ready for review when it is coherent and source-supported with open assumptions visible. It is ready for baseline only when relevant questions are resolved, both reviews are complete and formal sign-off requirements are met.

5.4 Meetings, decisions and conflict resolution

Keep a short record for each actual meeting: date, attendees, artefacts examined, decisions, open questions and actions with owners/target dates. Link to issues or decision entries where practical. Do not record meetings that have not occurred.

For disagreements, compare the options against stakeholder value, required scope, evidence, capacity, quality, security and risk. Seek a reasoned agreement; record genuine significant decisions with rationale and consequences in Willem's Decision Log. If evidence is insufficient, record a justified deferment and the information needed. Ask the lecturer to resolve unclear assessment requirements; do not present internal assumptions as lecturer approval [S4, sections 12-14].

5.5 Delays and changes in availability

Report a likely missed task or review commitment promptly, identify completed work and explain the remaining dependency. Agree and record a revised handoff or redistribution. Track significant schedule variance and its engineering cause. Do not silently cut required security, testing or peer review to recover time [S4, section 18].

Each member should retain enough familiarity with the other artefacts to help resolve blockers and answer assessor questions. Reassignment does not justify invented contribution or attendance evidence.

5.6 AI, references and sensitive information

Each author remains accountable for material AI-assisted work. Record date, student, tool, engineering task, contribution, verification, decision and issues found in the team AI Usage Register. Verify important claims against credible sources or appropriate tests and be able to explain and modify the submitted work [S4, sections 10-10.1].

Use in-text citations and a reference list where claims depend on supplied or external material. Do not commit credentials or expose confidential/sensitive information to external

AI systems. Actual repository configuration claims require project evidence, not a citation to this agreement [S4, sections 9-10; S1, section 9.1].

5.7 Acceptance and revision

All three members confirmed the Team Working Agreement and approved its use for Milestone 1 on 8 September 2026.

Member	Agreement status	Confirmation date
Ryno Goetz	Agreed	Confirmed and approved, 8 September 2026
Steven Riaan Piek	Agreed	Confirmed and approved, 8 September 2026
Willem Booysen	Agreed	Confirmed and approved, 8 September 2026

Review this agreement if availability or project conditions change. Controlled amendments require the same meaningful review and approval process.

References: [Source register](#), S4 sections 7.1-10, 12-14, 18-19; S1 sections 3, 7, 9.1 and 10; S2 and S3.

6. GitHub governance and team controls

6.1 Repository and mandatory controls

The local repository's configured remote is RynoGz/SEN831_Group_F. Its tracked layout currently includes the README and Milestone_1 through Milestone_4 folders. Ryno confirmed that all three members have full repository access. This does not prove team registration or that branch controls are enabled.

GOV-001

One controlled team repository unless an alternative is justified and approved.

Governing source: S4 section 9.

GOV-002

Protect main and treat it as the controlled product state.

Governing source: S4 section 9.

GOV-003

No substantive direct development on main; use PRs for substantive changes entering it.

Governing source: S4 section 9.

GOV-004

At least two approvals from team members other than the author; self-approval is not accepted.

Governing source: S4 section 9.

GOV-005

Reviews must meaningfully inspect the work.

Governing source: S4 sections 9-9.1.

GOV-006

Meaningful engineering work appears in a project board/backlog and is linked where practical.

Governing source: S4 section 9.

GOV-007

Do not commit passwords, keys, tokens or confidential credentials.

Governing source: S4 section 9.

GOV-008

Commits, PRs, reviews, merges and changes must be progressive and authentic.

Governing source: S4 sections 9 and 23.

The main branch is protected and the team's controlled review policy requires both non-author members to approve substantive changes. The final M1 revision received both required reviews and approvals.

6.2 Author and reviewer responsibilities

Ryno Goetz

Steven Riaan Piek and Willem Booysen.

Steven Riaan Piek

Ryno Goetz and Willem Booysen.

Willem Booysen

Ryno Goetz and Steven Riaan Piek.

For Ryno's analysis, Steven should particularly check whether needs/scope produce testable, traceable requirements. Willem should particularly check risks, assumptions and decision consequences. Both reviewers inspect the complete change before approving; the focus areas do not divide their responsibility into separate partial approvals.

6.3 Document workflow

1. Represent a coherent task in the backlog, such as reviewing the requester/staff/management stakeholder analysis.
2. Create a task branch from the latest approved main state.
3. Edit Markdown source files and verify source citations, IDs, linked artefacts and declared assumptions.
4. Open a PR linked to the issue, describing the engineering purpose, affected artefacts, verification and unresolved questions.
5. Request both non-author reviewers, address findings and have the corrected content re-reviewed as necessary.
6. Merge only after the final content satisfies the required approvals and applicable controls. Link the result to the issue and update dependent work.

6.4 Task and pull request information

An issue should identify its owner, output, source/requirement links, dependencies and completion criteria. Record an agreed target date when available. A PR should link the issue and explain why the change is needed, what it affects, what was actually checked, what remains open and how material AI assistance was verified.

For documentation review, check at least these relevant questions: Does the statement match its source? Is a proposal being mistaken for an approved fact? Does scope preserve every required capability? Are identifiers consistent? Does this change affect requirements, risks or future verification? Are approvals based on the final content? For code in later milestones, extend review to implementation correctness, tests/regression, dependencies, maintainability and security [S4, section 9.1 and Appendix A].

6.5 Governance status

Main protection and review settings

Main is protected, and the team applies the Master Brief requirement of two non-author approvals.

Member access/registration

Ryno, Steven and Willem have full repository access.

Issue, branch and pull request

The M1 document change was managed through the team repository workflow.

Two independent reviews of Ryno's change

Both non-author members completed meaningful review and approval.

Controlled merge/version

PED v1.0 records the approved M1 baseline; later revisions follow the controlled change workflow.

Repository history remains the authoritative record of controlled work. M1 documentation provides valid engineering evidence before application coding or a CI pipeline is implemented [S4, section 23; S1, sections 3.1 and 5].

7. Requirements and acceptance criteria

7.1 Purpose

This document defines the baselined functional requirements (FR) and non-functional requirements (NFR) for CivicConnect for Milestone 1.

The requirements are derived from:

- the SEN381 CivicConnect Master Project Brief;
- Ryno Goetz's Member 1 stakeholder, scope, constraints and assumptions artefacts, version 0.2;
- the stakeholder needs (NEED-*) and scope capabilities (SCP-*) established by Member 1;
- requirements-engineering guidance from ISO/IEC/IEEE 29148:2018; and
- product-quality guidance from ISO/IEC 25010:2023.

The team reviewed the Member 1 foundation together with the requirements and RTM. The requirements in this PED form part of the approved v1.0 baseline.

7.2 Requirement conventions

Field	Meaning
ID	Stable requirement identifier (FR-* or NFR-*)
Source / stakeholder	Primary source and stakeholder(s) affected
Need / scope	Traceability to Member 1 NEED-* and SCP-*
Priority	Must = required minimum capability; Should = important cross-cutting quality obligation
Status	Baselined in PED v1.0; changes require control

The wording uses shall for mandatory requirements. Open policy questions are not silently answered where the Member 1 artefacts identify them as unresolved.

7.3 Functional requirements

FR-001 — Submit a service request

Source / stakeholders: S4 section 3; STK-001 Requesters; STK-002 Authorised service staff

Need / scope: NEED-001 / SCP-001

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall permit a requester to submit a new service request with a title, description, category, location, requester identity/contact information, and a sensitive-information indicator. The system shall automatically record the submission date/time and allow an optional attachment.

Acceptance Criteria:

1. A submission is rejected when any required field is missing.
2. A successfully submitted request contains all required request information.
3. The submission date/time is recorded automatically rather than entered manually by the requester.
4. Omitting an attachment does not prevent a valid request from being submitted.
5. After successful submission, the request is recorded in the Submitted state.
6. The M1 capability does not require live ingestion from WhatsApp, email, telephone, or paper workflows.

FR-002 — Controlled request categorisation

Source / stakeholders: S4 section 3; STK-001 Requesters; STK-004 Service coordination

Need / scope: NEED-002 / SCP-002

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall provide a controlled category mechanism using the initial categories defined in the Member 1 working scope baseline: Maintenance, IT support, Facility fault, Damaged equipment, Security concern, Lost property, and Other.

Acceptance Criteria:

1. A requester selects a category from the available controlled category set.
2. A submitted request stores the selected category.
3. The initial category set contains the seven categories defined in the Member 1 working scope baseline.
4. A request cannot use an uncontrolled free-text category value instead of a controlled category.
5. The M1 requirement does not require AI-based automatic classification.

FR-003 — View current request status

Source / stakeholders: S4 section 3; STK-001 Requesters

Need / scope: NEED-003 / SCP-003

Priority: Must

Status: Baselined in PED v1.0

Requirement:

An authorised requester shall be able to view the status of their submitted service requests through CivicConnect.

Acceptance Criteria:

1. An authorised requester can view the status of a request they are permitted to access.
2. The displayed status matches the recorded lifecycle state of the request.
3. A requester cannot access another request through the status view without authorisation.
4. Status values shown to the requester use the controlled lifecycle terminology.

FR-004 — View request history/list

Source / stakeholders: S4 section 3; STK-001 Requesters

Need / scope: NEED-003 / SCP-004

Priority: Must

Status: Baselined in PED v1.0

Requirement:

Upon confirmation of the retention policy, CivicConnect shall permit an authorised requester to view a retrievable list/history of their previously submitted service requests.

Acceptance Criteria:

1. An authorised requester can retrieve requests they are permitted to view.
2. The history identifies each request sufficiently to distinguish it from other requests.
3. Historic requests preserve their recorded lifecycle information.
4. The system does not claim infinite storage; the final retention period remains subject to confirmation.

FR-005 — Provide in-application request feedback

Source / stakeholders: S4 section 3; STK-001 Requesters

Need / scope: NEED-003 / SCP-005

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall provide the requester with meaningful in-application feedback when a request is accepted, rejected, updated, resolved, or closed.

Acceptance Criteria:

1. The requester can identify when their request has been accepted.
2. The requester can identify when their request has been rejected.
3. The requester can identify relevant updates to their request.
4. The requester can identify when their request is resolved.
5. The requester can identify when their request is closed.

6. The M1 requirement does not require SMS, WhatsApp, or email notification delivery.

FR-006 — View work relevant to authorised staff

Source / stakeholders: S4 section 3; STK-002 Authorised service staff

Need / scope: NEED-004, NEED-009 / SCP-006

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall allow authorised service staff to access service requests consistent with their duties and permissions.

Acceptance Criteria:

1. An authorised staff user can view requests within the scope of their permitted work.
2. A staff member cannot automatically view all requests merely because they are staff.
3. Access decisions are based on the agreed responsibility and permission model.
4. Sensitive requests remain subject to the applicable access rules.

FR-007 — Search, filter and sort requests

Source / stakeholders: S4 section 3; STK-002 Authorised service staff

Need / scope: NEED-004 / SCP-007

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall permit authorised staff to search, filter, and/or sort service requests according to useful criteria agreed for the operational workflow.

Acceptance Criteria:

1. An authorised staff member can retrieve relevant requests using the agreed search, filter, and/or sort capabilities.
2. Search and filtering apply only to requests the user is authorised to access.
3. The returned results match the selected search, filter, or sort criteria.
4. Exact operational criteria are confirmed during requirements review rather than invented as an M1 fact.
5. A sophisticated search engine is not required unless the team justifies it.

FR-008 — View authorised request details

Source / stakeholders: S4 section 3; STK-002 Authorised service staff

Need / scope: NEED-004, NEED-009 / SCP-008

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall allow an authorised user to view request details within the scope of their responsibility and access rights, including sensitive information only where authorised.

Acceptance Criteria:

1. An authorised user can view request information permitted by their role or responsibility.
2. Sensitive information is not displayed to a user without the required authorisation.

3. The detailed view includes the request information captured by FR-001, subject to the user's permissions.
4. Administrative rights do not automatically grant access to business request details.

FR-009 — Assign, accept and reassign responsibility

Source / stakeholders: S4 sections 2–3; STK-002 Authorised service staff; STK-004 Service coordination

Need / scope: NEED-005 / SCP-009

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall allow authorised staff to accept or be assigned responsibility for a service request, and shall allow authorised coordinators or managers to assign or reassign responsibility.

Acceptance Criteria:

1. An authorised staff user can accept responsibility where permitted.
2. An authorised coordinator or manager can assign responsibility to an appropriate staff user.
3. An authorised coordinator or manager can reassign responsibility where permitted.
4. A user without the required permission cannot assign or reassign responsibility.
5. The current responsible user or responsibility state is identifiable for the request.

FR-010 — Enforce controlled request status transitions

Source / stakeholders: S4 section 3; STK-002 Authorised service staff; STK-004 Service coordination

Need / scope: NEED-006 / SCP-010

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall enforce the working request lifecycle Submitted → Accepted → Assigned → In Progress → Resolved → Closed, with controlled rejection from Submitted, authorised reassignment, and a controlled mechanism for requesting reopening.

Acceptance Criteria:

1. A newly submitted request starts in the Submitted state.
2. A request can transition only through transitions permitted by the agreed lifecycle rules.
3. An authorised user can reject a request only when it is in the Submitted state and the user's permissions allow rejection.
4. Assignment and reassignment are permitted only for authorised users.
5. A status change records the user responsible for the change.
6. An unauthorised or invalid status transition is blocked.
7. Reopening follows the approved working policy recorded under Q-003; later refinements require controlled change.

FR-011 — Record actions, comments and resolution information

Source / stakeholders: S4 section 3; STK-002 Authorised service staff

Need / scope: NEED-006 / SCP-011

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall permit authorised staff to record relevant actions, comments, and resolution information against a service request, with recorded information controlled for its intended visibility.

Acceptance Criteria:

1. An authorised user can record a relevant action or comment against a request.
2. An authorised user can record resolution information when resolving work.
3. Recorded entries remain associated with the correct request.
4. Where the final policy requires it, the system distinguishes requester-visible information from information restricted to authorised users.
5. Accountable actions have a responsible user attached to them.

FR-012 — Resolve or close requests when authorised

Source / stakeholders: S4 section 3; STK-002 Authorised service staff; STK-004 Service coordination

Need / scope: NEED-006 / SCP-012

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall permit only authorised users to resolve or close service requests in accordance with the controlled request lifecycle and agreed responsibilities.

Acceptance Criteria:

1. An authorised user can move a request to Resolved when the agreed resolution conditions are met.
2. An authorised user can move a request to Closed when the agreed closure conditions are met.
3. An unauthorised user cannot resolve or close a request.
4. The system records the user responsible for the resolution or closure action.
5. The approved working distinction between Resolved and Closed applies; later refinements to reopening rules require controlled change.

FR-013 — View service activity

Source / stakeholders: S4 section 3; STK-003 Management/oversight

Need / scope: NEED-007 / SCP-013

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall provide authorised management/oversight users with sufficient service activity information to monitor the request workflow.

Acceptance Criteria:

1. An authorised management/oversight user can view organisational service activity within their permitted scope.
2. The activity information includes recorded requests and relevant lifecycle information.
3. Access to the activity view is restricted according to the agreed permission model.
4. The management view is validated against representative request scenarios during later design and testing.

FR-014 — Identify open, overdue, resolved and closed work

Source / stakeholders: S4 section 3; STK-003 Management/oversight

Need / scope: NEED-007 / SCP-014

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall permit authorised users to identify open, overdue, resolved, and closed service requests using the agreed lifecycle and overdue definitions.

Acceptance Criteria:

1. Requests in the relevant open states can be identified separately from resolved, closed, and rejected requests.
2. Open work can be identified from the controlled lifecycle state.
3. A request is considered overdue when its agreed target date/time has passed and its status is not Resolved, Closed, or Rejected.
4. The target date/time is set by an authorised staff member or manager according to the working policy.
5. Overdue classification is verified using a controlled test case at or around the target date/time boundary.

FR-015 — Analyse requests by category, status and justified dimensions

Source / stakeholders: S4 section 3; STK-003 Management/oversight

Need / scope: NEED-008 / SCP-015

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall permit authorised users to analyse requests by category and status, and, where justified, by other agreed dimensions.

Acceptance Criteria:

1. Authorised management users can group, filter, or otherwise present request information by category.
2. Authorised management users can analyse request information by lifecycle status.
3. Any additional analysis dimension is included only when its operational value is justified and agreed.
4. The analysis respects the visibility permissions of the authorised user.
5. Predictive analytics are not required for M1.

FR-016 — Support accountability and service-performance analysis

Source / stakeholders: S4 section 3; STK-003 Management/oversight

Need / scope: NEED-008 / SCP-016

Priority: Must

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall retain sufficient information about attributable requests, responsibilities, and progress to support accountability and service-performance analysis.

Acceptance Criteria:

1. The system retains sufficient lifecycle history to determine how a request progressed.
2. Responsibility for relevant work can be identified from recorded request information.
3. Status changes can be attributed to the user who made them.
4. Management can explain request progress and responsibility using recorded information.
5. Predictive analytics and a data warehouse are not required for M1.

7.4 Non-functional requirements

NFR-001 — Authorisation and least-privilege access

Source / stakeholders: S4 sections 2, 4 and 16; STK-001, STK-002, STK-003, STK-005

Need / scope: NEED-009 / SCP-017

Priority: Should

Status: Baselined in PED v1.0

Requirement:

Using the least-privilege approach, CivicConnect shall limit request access and permitted actions according to authorised responsibilities.

Acceptance Criteria:

1. A requester can view their own requests in accordance with the agreed visibility policy.
2. Staff can view work relating to their authorised role or responsibility.
3. Only authorised users can assign, reassign, change status, resolve, or close requests.
4. Access to management reporting is permission-controlled.
5. Administrative privileges do not automatically grant business-data access.

NFR-002 — Sensitive-information protection

Source / stakeholders: S4 sections 2, 4 and 16; STK-001, STK-002, STK-005

Need / scope: NEED-009 / SCP-017

Priority: Should

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall prevent unauthorised disclosure of sensitive request information throughout the request lifecycle.

Acceptance Criteria:

1. Sensitive information is identified using the agreed sensitive-information indicator and protected according to the applicable access rules.

2. Only authorised users can view sensitive request information.
3. Unauthorised attempts to access protected information are denied.
4. Verification uses fictional, non-sensitive test data.
5. The final retention period is not invented in M1 and remains subject to confirmation.

NFR-003 — Auditability and accountability

Source / stakeholders: S4 sections 3, 8 and 11; STK-002, STK-003, STK-006

Need / scope: NEED-006, NEED-008 / SCP-010, SCP-011, SCP-016

Priority: Should

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall maintain attributable records of relevant lifecycle changes, responsibility changes, and actions so that request history can be reviewed for accountability purposes.

Acceptance Criteria:

1. Each status change identifies the resulting status and responsible user.
2. Assignments and reassignments are attributable to the user who performed them.
3. Relevant actions, comments, and resolution information remain associated with the request.
4. The resulting history is sufficient to reconstruct how the request progressed during verification.

NFR-004 — Usability and understandable feedback

Source / stakeholders: S4 sections 3 and 15; STK-001, STK-002, STK-003

Need / scope: NEED-001, NEED-003, NEED-004 / SCP-001, SCP-003 to SCP-007, SCP-013 to SCP-016

Priority: Should

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall present request information, status, and feedback in a form that the intended user can understand and use for the supported task.

Acceptance Criteria:

1. Requesters can identify the current status of their own requests during representative task-based verification.
2. Feedback identifies the relevant request event or outcome.
3. Staff can identify relevant work by following an agreed search, filter, and/or sort workflow.
4. Management can interpret the status and activity information used for oversight.
5. Usability is supported by defined verification tasks and criteria rather than an unsupported claim that the system is “user-friendly.”

NFR-005 — Data integrity and consistency

Source / stakeholders: S4 sections 3, 11 and 15; STK-002, STK-003

Need / scope: NEED-001, NEED-005, NEED-006, NEED-008 / SCP-001, SCP-009 to SCP-016

Priority: Should

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall maintain consistency between recorded request details, responsibility, lifecycle state, and attributable history.

Acceptance Criteria:

1. Each request has one current lifecycle state at a time.
2. Changes in responsibility are recorded consistently.
3. The status history reflects the status changes that occurred.
4. An unauthorised or invalid transition does not leave the request in an inconsistent lifecycle state.
5. Repeated and boundary cases are included in later verification.

NFR-006 — Performance under agreed operating conditions

Source / stakeholders: S4 sections 15 and 18; STK-003, STK-005, STK-006

Need / scope: NEED-010 / SCP-018

Priority: Should

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall support responsive request submission, retrieval, search, filtering, sorting, and oversight operations under the agreed operating conditions and workload.

Acceptance Criteria:

1. Working M1 performance target: using 1,000 seeded service requests and three concurrent representative users, at least 95% of request retrieval, search, filter, sort and management-view operations complete within two seconds.
2. Under the same test conditions, at least 95% of valid request submissions complete and return acknowledgement within three seconds.
3. Verification evidence records the environment, dataset, concurrency, operation, sample count, measured response times and pass/fail result.
4. The team reviews these working targets during architecture and deployment planning; any approved change is recorded through change control.

NFR-007 — Reliability and recoverability

Source / stakeholders: S4 sections 15, 17 and 18; STK-005, STK-006

Need / scope: NEED-010 / SCP-018

Priority: Should

Status: Baselined in PED v1.0

Requirement:

CivicConnect shall support controlled recovery of service-request information and configuration appropriate to the agreed operating environment.

Acceptance Criteria:

1. Service-request records, status history, assignments and required configuration are included in the defined backup scope.
2. A recovery test restores the agreed backup into a controlled environment within 30 minutes.

3. The recovery test demonstrates a maximum planned data-loss window of 24 hours and records the backup time, restore time, environment and result.
4. The 30-minute recovery and 24-hour data-loss targets are working M1 thresholds and must be reviewed against the selected platform before the v2.0 architecture baseline.

NFR-008 — Maintainability and controlled change

Source / stakeholders: S4 sections 6, 8, 11 and 18; STK-006 Project team

Need / scope: NEED-010, NEED-011 / SCP-018

Priority: Should

Status: Baselined in PED v1.0

Requirement:

Using requirement identifiers, traceability, version history, and verification evidence, CivicConnect and its engineering artefacts shall support controlled change.

Acceptance Criteria:

1. Requirements retain stable identifiers unless a controlled change explicitly requires a new identifier.
2. Approved requirement changes are reflected in the RTM and affected artefacts.
3. Design, implementation, and test evidence can later be associated with the applicable requirement.
4. Material changes follow the team-controlled change process.
5. Repository history records substantial changes without silently overwriting approved evidence.

7.5 Open questions and controlled deferrals

The following client-policy items remain controlled deferrals because the supplied briefs do not determine them. Any later resolution must follow change control.

Open question	Affected requirements	Current control
Q-001: Exact required request information, categories, and who may change them	FR-001, FR-002, FR-007	Working policy approved for v1.0; refine through controlled change if stakeholder evidence differs
Q-002: Final roles, visibility, assignment/reassignment, resolve/close authority	FR-006, FR-008, FR-009, FR-012, NFR-001	Least-privilege policy approved for v1.0; refine the permission matrix through controlled change
Q-003: Detailed state-transition, rejection, resolved/closed, and reopening policy	FR-010, FR-012	Lifecycle policy approved for v1.0; refine detailed transitions through controlled change
Q-004: Final overdue definition, time basis, and state treatment	FR-014	Working overdue definition approved; verify boundary cases during later design and testing
Q-005: Final feedback timing, content, and mechanism	FR-005	In-application feedback approved; refine content and timing through controlled change
Q-006: Sensitive-information handling and retention policy	FR-004, FR-008, NFR-002	Least privilege and non-sensitive test data approved; retention remains a controlled deferral

Open question	Affected requirements	Current control
Q-007: User volumes, operating hours, devices/connectivity, and other conditions needed to define measurable quality criteria	NFR-004, NFR-006, NFR-007	M1 performance and recovery targets approved; refine operating conditions through controlled change

The M1 performance and recovery targets baseline NFR-006 and NFR-007. Client-specific workload, retention and operating constraints may refine them later only through controlled review.

7.6 Requirements quality basis

The requirements are intended to be clear, uniquely identifiable, traceable to sources/stakeholders, and verifiable. ISO/IEC/IEEE 29148:2018 provides the requirements-engineering basis for requirements processes and information items. ISO/IEC 25010:2023 provides the product-quality model used to structure the cross-cutting quality requirements.

The project Master Brief remains the authoritative source for project-specific requirements and controls. The ISO standards are supporting engineering references and do not override the assignment brief.

8. Requirements traceability matrix

8.1 Purpose

This Requirements Traceability Matrix (RTM) provides the controlled link between the Member 1 stakeholder/source analysis, stakeholder needs, scope capabilities, requirements, and acceptance criteria.

Later lifecycle evidence is intentionally marked TBD because M1 does not yet contain the final architecture, implementation, tests, or release evidence.

The intended lifecycle trace is:

Stakeholder/Source → Need → Scope Capability → Requirement → Acceptance Criteria → Design → Issue/PR → Implementation → Test → Acceptance/Release Evidence

8.2 Traceability matrix

Source / Stakeholder	Need	Scope	Requirement	Acceptance criteria	Design	Issue / PR	Implementation	Test	Acceptance / Release
STK-001 / S4	NEED-001	SCP-001	FR-001	FR-001 AC1–AC6	TBD	TBD	TBD	TBD	TBD
STK-001, STK-004 / S4	NEED-002	SCP-002	FR-002	FR-002 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-001 / S4	NEED-003	SCP-003	FR-003	FR-003 AC1–AC4	TBD	TBD	TBD	TBD	TBD
STK-001 / S4	NEED-003	SCP-004	FR-004	FR-004 AC1–AC4	TBD	TBD	TBD	TBD	TBD
STK-001 / S4	NEED-003	SCP-005	FR-005	FR-005 AC1–AC6	TBD	TBD	TBD	TBD	TBD
STK-002 / S4	NEED-004, NEED-009	SCP-006	FR-006	FR-006 AC1–AC4	TBD	TBD	TBD	TBD	TBD
STK-002 / S4	NEED-004	SCP-007	FR-007	FR-007 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-002 / S4	NEED-004, NEED-009	SCP-008	FR-008	FR-008 AC1–AC4	TBD	TBD	TBD	TBD	TBD
STK-002, STK-004 / S4	NEED-005	SCP-009	FR-009	FR-009 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-002, STK-004 / S4	NEED-006	SCP-010	FR-010	FR-010 AC1–AC7	TBD	TBD	TBD	TBD	TBD
STK-002 / S4	NEED-006	SCP-011	FR-011	FR-011 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-002, STK-004 / S4	NEED-006	SCP-012	FR-012	FR-012 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-003 / S4	NEED-007	SCP-013	FR-013	FR-013 AC1–AC4	TBD	TBD	TBD	TBD	TBD
STK-003 / S4	NEED-007	SCP-014	FR-014	FR-014 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-003 / S4	NEED-008	SCP-015	FR-015	FR-015 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-003 / S4	NEED-008	SCP-016	FR-016	FR-016 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-001, STK-002, STK-003, STK-005 / S4	NEED-009	SCP-017	NFR-001	NFR-001 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-001, STK-002, STK-005 / S4	NEED-009	SCP-017	NFR-002	NFR-002 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-002, STK-003, STK-006 / S4	NEED-006, NEED-008	SCP-010, SCP-011, SCP-016	NFR-003	NFR-003 AC1–AC4	TBD	TBD	TBD	TBD	TBD
STK-001, STK-002, STK-003 / S4	NEED-001, NEED-003, NEED-004	SCP-001, SCP-003–SCP-007, SCP-013–SCP-016	NFR-004	NFR-004 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-002, STK-003 / S4	NEED-001, NEED-005, NEED-006, NEED-008	SCP-001, SCP-009–SCP-016	NFR-005	NFR-005 AC1–AC5	TBD	TBD	TBD	TBD	TBD
STK-003, STK-005, STK-006 / S4	NEED-010	SCP-018	NFR-006	NFR-006 AC1–AC4	TBD	TBD	TBD	TBD	TBD
STK-005, STK-006 / S4	NEED-010	SCP-018	NFR-007	NFR-007 AC1–AC4	TBD	TBD	TBD	TBD	TBD
STK-006 / S4	NEED-010, NEED-011	SCP-018	NFR-008	NFR-008 AC1–AC5	TBD	TBD	TBD	TBD	TBD

8.3 Requirement-to-source coverage

Requirement group	Covered source capabilities
FR-001	SCP-001 / NEED-001
FR-002	SCP-002 / NEED-002
FR-003	SCP-003 / NEED-003
FR-004	SCP-004 / NEED-003
FR-005	SCP-005 / NEED-003
FR-006	SCP-006 / NEED-004, NEED-009
FR-007	SCP-007 / NEED-004
FR-008	SCP-008 / NEED-004, NEED-009
FR-009	SCP-009 / NEED-005
FR-010	SCP-010 / NEED-006
FR-011	SCP-011 / NEED-006
FR-012	SCP-012 / NEED-006
FR-013	SCP-013 / NEED-007
FR-014	SCP-014 / NEED-007
FR-015	SCP-015 / NEED-008
FR-016	SCP-016 / NEED-008
NFR-001, NFR-002	SCP-017 / NEED-009
NFR-003	SCP-010, SCP-011, SCP-016 / NEED-006, NEED-008
NFR-004	SCP-001, SCP-003–SCP-007, SCP-013–SCP-016
NFR-005	SCP-001, SCP-009–SCP-016
NFR-006, NFR-007, NFR-008	SCP-018 / NEED-010, NEED-011

8.4 Open-question traceability

Open question	Affected requirements	Current control
Q-001 — Exact request information, categories, and category-change authority	FR-001, FR-002, FR-007	Working policy approved for v1.0; refine through controlled change if stakeholder evidence differs
Q-002 — Final roles, visibility, assignment/reassignment, resolve/close authority	FR-006, FR-008, FR-009, FR-012, NFR-001	Least-privilege policy approved for v1.0; refine the permission matrix through controlled change
Q-003 — Detailed transitions, rejection, resolved/closed, and reopening	FR-010, FR-012	Lifecycle policy approved for v1.0; refine detailed transitions through controlled change
Q-004 — Overdue time basis and state treatment	FR-014	Working overdue definition approved; verify boundary cases during later design and testing
Q-005 — Feedback timing, content, and mechanism	FR-005	In-application feedback approved; refine content and timing through controlled change
Q-006 — Sensitive-information handling and retention	FR-004, FR-008, NFR-002	Least privilege and non-sensitive test data approved; retention remains a controlled deferral

Open question	Affected requirements	Current control
Q-007 — User volumes, operating hours, devices/connectivity, and other conditions needed for measurable quality criteria	NFR-004, NFR-006, NFR-007	M1 performance and recovery targets approved; refine operating conditions through controlled change

8.5 Later-lifecycle evidence

The following columns remain intentionally open in M1:

- Design: populated when architecture/design decisions are made.
- Issue / PR: link the issue and PR that implement or change the requirement.
- Implementation: link the relevant code, configuration, or documentation evidence.
- Test: link test cases/results demonstrating the acceptance criteria.
- Acceptance / Release: link approved baseline/release evidence.

No later evidence is claimed before it exists.

8.6 Traceability control rules

- Requirement IDs remain stable unless a controlled change explicitly requires otherwise.
- Any approved requirement change updates this RTM and affected artefacts.
- Acceptance criteria remain linked to the requirement they verify.
- A requirement must not be marked complete merely because it has been designed or implemented; verification evidence is required.
- Later design, implementation, and test evidence must link back to the applicable requirement ID.
- Open questions remain visible until resolved through the team's change/review process.
- The RTM forms part of the evolving PED and must preserve its version history.

8.7 Status summary

Category	Count	Status
Functional requirements	16	Baselined
Non-functional requirements	8	Baselined
Total requirements	24	Baselined
Scope capabilities SCP-001–SCP-016	16	Covered
Cross-cutting obligations SCP-017–SCP-018	2	Covered
Later design/implementation/test/release evidence	—	TBD

9. Initial risk register

The register records current M1 risks. Owners coordinate treatment but do not remove collective responsibility. Status remains Open until the mitigation is evidenced or the risk is formally closed.

ID	Description	Cause	Consequence	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
R1	Scope creep from new stakeholder demands	Features requested after baseline	Delay, missed deadlines and increased cost	Medium	High	High	Defend baseline and perform impact analysis	Reprioritise or formally adjust scope and schedule	Willem Booysen	Open
R2	GitHub governance not enforced	Protection or review controls incomplete	Uncontrolled changes and weak traceability	Medium	High	High	Require protected main and two non-author approvals	Audit settings and stop non-compliant merges	Ryno Goetz	Open
R3	Security requirements overlooked	Sensitive-data and access rules remain incomplete	Disclosure, redesign or compliance failure	Medium	High	High	Baseline least-privilege and sensitive-data requirements	Security review and corrective change request	Steven Riaan Piek	Open
R4	Forward-engineering concerns ignored	Later concerns are deferred without criteria	Technical debt and constrained design options	Medium	Medium	Medium	Maintain forward-considerations and decision evidence	Revisit before M2 decisions	Willem Booysen	Open
R5	Weak PED document control	Version history or controlled changes become incomplete	Assessment penalty and an unreliable baseline	Low	High	Medium	Maintain version history and require team review for controlled changes	Issue a corrected version and repeat the affected review	All members	Open
R6	Schedule slippage	Effort underestimated or feedback delayed	Rushed work and missed milestone evidence	Medium	High	High	Daily tracking, review capacity and buffer time	Reallocate work and control scope	Ryno Goetz	Open
R7	Team governance breakdown	Poor accountability or collaboration	Missed artefacts and weak individual evidence	Medium	Medium	Medium	Working agreement, issues and regular check-ins	Escalate unresolved breakdown to lecturer	All members	Open
R8	Deployment environment mismatch	Client infrastructure constraints are unknown	Rework or failed deployment	Medium	High	High	Confirm environment constraints before stack selection	Use a compatible alternative and update ADR	Willem Booysen	Open
R9	Dependency or service-limit failure	External libraries or free tiers are not evaluated	Build failure, vulnerability or unexpected cost	Medium	High	High	Record dependencies, maturity, limits and likely cost	Replace, patch or select an approved alternative	All members	Open
R10	AI accountability failure	Material AI use is incomplete or insufficiently verified	Integrity penalty or indefensible artefact	Low	High	High	Maintain required AI fields and retain human verification	Correct or withdraw unsupported material	All members	Open

9.1 Highest concern risk

R1 remains the highest immediate concern because it can trigger schedule, quality, cost and review failures simultaneously. R2, R3, R6, R8, R9 and R10 also carry High priority. R1 is selected as the lead concern by tie-break because stakeholder pressure can arise immediately after baseline and can create cascading effects across several other risks. The response is to defend the agreed scope and require formal impact analysis before accepting additions [S1, section 3; S4, sections 4, 12 and 14].

10. Forward engineering considerations

Concern	Why it matters now	Later activity affected	Missing information	Risk of ignoring
Security	Protects sensitive request information and access boundaries	Architecture, identity and deployment	Applicable privacy rules and final permission matrix	Costly redesign or disclosure
Testability	Requirements and acceptance criteria must support verification	Test strategy, automation and CI	Architecture, stack and risk-based test priorities	Late defects and weak evidence
Deployment environment	Environment limits can constrain technology	Hosting and production-like staging	Client infrastructure, connectivity and support constraints	Rework or deployment failure
Data migration and recovery	Existing fragmented records may need controlled treatment	Persistence, migration, backup and restore	Source formats, quality, retention and ownership	Data loss, delay or failed migration
Observability	Failures and overdue work must be detectable	Logging, monitoring and incident response	Operational metrics, alert ownership and tooling	Failures remain hidden
Maintainability	A small team must support controlled future change	Modularity, standards and technical-debt control	Final architecture and team capability evidence	Costly change and accumulated debt
Operational cost	The solution must remain sustainable beyond the classroom	Platform, scaling and support decisions	Service pricing, free-tier limits and expected usage	Unsustainable or locked-in solution

These seven concerns preserve later options without selecting a final stack, architecture, database, testing framework or deployment platform in M1 [S1, sections 4-5; S4, sections 15-18].

11. Engineering decision log

ID	Context and constraints	Alternatives	Decision and rationale	Trade-offs and risks	Evidence	Deferred?	Later consequence
D1	Three-person team, M1 schedule and mandatory web workflow	Include mobile application scope now	Exclude native mobile applications from the initial baseline; web delivery covers the minimum capabilities with lower workload	Less mobile-specific access; avoids duplicated design, testing and deployment	Scope baseline and stakeholder review	No	Reassess only through change control
D2	The Master Brief requires protected main, Pull Requests and two non-author approvals	One approval; no branch protection	Apply protected main and require both non-author members to approve substantive changes	Review takes time but strengthens accountability and baseline integrity	Protected-main configuration and completed M1 reviews	No	Every substantive author depends on both other reviewers
D3	The integrated PED, reviews and M1 registers are complete	Retain separate member reports; delay approval	Approve the unified PED v1.0 as the controlled M1 engineering baseline	Creates a stable foundation; later changes require formal control	Version history, team review and approved PED	No	M2 work must use this baseline and record approved changes
D4	M1 is not a technology-selection milestone and client constraints are incomplete	Choose a familiar stack now	Defer final stack selection until M2 evidence is available	Preserves flexibility; delays implementation-specific planning	Requirements stability, capability, environment, security and cost research	Yes	M2 must compare viable alternatives
D5	Acceptance criteria exist, but stack, architecture and automation environment are not selected	Select a test framework now	Defer the framework choice, not testability or acceptance criteria	Avoids premature tool lock-in; requires M2 follow-up	Architecture, stack and risk-based test strategy	Yes	M2 maps test techniques and tools to requirements and risks
D6	Stakeholder needs may conflict and no single stakeholder has unlimited authority	Ignore conflict; escalate every disagreement	Use evidence-based negotiation and record significant unresolved decisions	Discussion consumes time; improves transparency and trust	Meeting notes, issue discussion and affected artefacts	No	Conflicts remain traceable to scope and requirements
D7	AI is permitted only with disclosure, verification and student accountability	Ban AI; accept AI output without verification	Allow material AI assistance when it is recorded and independently checked	Faster drafting but added verification and integrity risk	AI Usage Register and reviewed artefacts	No	Students must explain and modify all submitted work

12. AI usage register

Material AI assistance is recorded below together with the human verification, decisions and issues identified by each responsible student.

Date	Student	Tool	Engineering task	AI contribution	Human verification	Decision	Issues found
7-8 Sep 2026	Ryno Goetz	OpenAI Codex	Foundation and governance documentation	Initial drafting, restructuring and consistency review	Checked against S1/S4 and corrected project-specific assumptions and scope	Accepted with corrections	Removed unsupported completion claims and clarified controlled deferrals
8 Sep 2026	Ryno Goetz	OpenAI Codex	Table visualisation	Converted register data into readable standalone table layouts	Compared every row and field with the source register	Accepted after verification	Corrected hidden merged-cell text and layout
8 Sep 2026	Ryno Goetz	OpenAI Codex	Unified PED quality review	Compared all member parts and proposed corrections	Rechecked findings against S1/S4 and verified the final corrections	Accepted with corrections	Found register fields, count and cross-document inconsistencies
8 Sep 2026	Steven Riaan Piek	ChatGPT	Requirements wording	Suggested clearer requirement and acceptance-criteria wording	Compared wording with the minimum capabilities, Member 1 scope and S1/S4	Accepted with corrections	Improved clarity and testability; wording was manually corrected
8 Sep 2026	Steven Riaan Piek	ChatGPT	RTM organisation	Suggested matrix structure and traceability columns	Checked all FR/NFR, stakeholder, need, scope and acceptance links	Accepted with corrections	Identified missing later-lifecycle links, retained as TBD for future milestones
8 Sep 2026	Willem Booyesen	Copilot	Risk prioritisation	Suggested R1 as the highest immediate concern and drafted reasoning	Compared R1 with other High risks using probability, impact, immediacy and cascading effects	Accepted with corrections	R1 tied by priority; tie-break rationale added

References

S1

SEN381 Teaching Team. SEN381 CivicConnect Project Milestone 1 Engineering Foundation and Requirements Baseline. 2026.

S2

RynoGz/SEN831_Group_F team repository and README, reviewed as part of the Milestone 1 baseline.

S3

Team project-planning confirmations recorded by Ryno Goetz on 7 September 2026: team membership, availability, communication and working arrangements.

S4

SEN381 Teaching Team. CivicConnect Master Project Brief Community Service Request Management Platform. Version 1.1. 2026.

S5

ISO/IEC/IEEE 29148:2018. Systems and software engineering Life cycle processes Requirements engineering.

S6

ISO/IEC 25010:2023. Systems and software engineering Systems and software Quality Requirements and Evaluation Product quality model.